

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Electronics and Electrical Communication Engineering
2. **Subject Name:** Communication-I Laboratory. **L-T-P:** 0-0-3 **Credits:**2
3. **Pre-requisites:** NA
4. **Syllabus and reference books:**

Syllabus: 1. Design and implementation of amplitude modulation systems with envelope detection, and study its (a) signal handling, and (b) frequency response characteristics 2. Design and implementation of DSBSC modulation system with coherent demodulation and study its (a) signal handling, and (b) frequency response characteristics when carrier is suppressed (i) fully, (ii) partially. 3. Design and implementation of frequency modulation system with VCO based modulation. (a) Signal handling and frequency response characteristics . 4. Design and implementation of frequency demodulator using PLL and also by band pass filter (a) signal handling, and (b) frequency response characteristics 5. Design and implementation of pulse width modulation and demodulation (b) signal handling and (b) frequency response characteristics. 6. Design and implementations of pulse amplitude modulation and demodulation (b) signal handling and (b) frequency response characteristics. 7. Design and implementation of various line coding to digital data such as RZ, AMI, Manchester 8. Design of basic PN sequences with various properties (a) study of autocorrelation of PN sequence etc.

Reference Books:

- 1) Madhow, U. (2014). Introduction to Communication Systems. United Kingdom: Cambridge University Press.
- 2) An Introduction To Analog And Digital Communications. (2009). India: Wiley India Pvt. Limited.
- 3) Carlson, A. B., Crilly, P. B. (2010). Communication Systems. Japan: McGraw-Hill Higher Education.
- 4) Gallager RG. Principles of Digital Communication. Cambridge University Press; 2008.
- 5) Proakis, J. G., Salehi, M. (2008). Digital Communications. United States: McGraw-Hill.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Switching modulator	3=1*3
2.	Envelope detector	3=1*3
3.	DSB-SC modulation and demodulation	6=2*3
4.	Frequency modulation and demodulation	6=2*3
5.	Pulse width modulation and demodulation	3=1*3
5.	Pulse amplitude modulation and demodulation	3=1*3
7.	Analog communication receiver including AGC	3=1*3
8.	Line coding	6=2*3
9.	PN sequence generation	6=2*3
Total number of hours		39 lab hours